Experimental Design - Exam 1 (Take-home)

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Problem 1

In a study of starting salaries for assistant professors, five male assistant professors at each of three types of institutions were randomly polled and their starting salaries (in \$1000) were recorded. The summary statistics are:

	Public	Private	Church-affiliated
Mean	53.4	70.8	56.1
Variance	72.8	48.2	60.3

There are $n_i = 5$ observations per group ($a = 3$, $N = 15$).

(a) Type of experimental design and model

This is a single-factor completely randomized design (CRD) with one factor "institution type" at three levels. A fixed-effects one-way ANOVA model is

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, 3, \ j = 1, \ldots, 5,$$

where μ is the overall mean, τ_i is the effect of institution type i with the usual constraint $\sum_{i=1}^{3} \tau_i = 0$, and $\varepsilon_{ij} \sim \text{NID}(0, \sigma^2)$. Strictly speaking this is an *observational* study (professors were polled rather than randomly assigned to institution types), but we analyze it using the single-factor ANOVA framework as for a completely randomized design.

(b) Test for differences in average starting salaries

We test

$$H_0 : \mu_1 = \mu_2 = \mu_3 \quad \text{vs} \quad H_a : \text{at least one } \mu_i \text{ differs.}$$

The grand mean is

$$\bar{y}_{..} = \frac{53.4 + 70.8 + 56.1}{3} = 60.1.$$

The treatment sum of squares is

$$SS_{\text{Trt}} = n \sum_{i=1}^{3} (\bar{y}_i - \bar{y}_{..})^2 = 5\left[(53.4 - 60.1)^2 + (70.8 - 60.1)^2 + (56.1 - 60.1)^2\right] = 876.9.$$

The error sum of squares is

$$SS_E = \sum_{i=1}^{3}(n_i - 1)s_i^2 = 4(72.8 + 48.2 + 60.3) = 725.2,$$

so the total sum of squares is

$$SS_T = SS_{\text{Trt}} + SS_E = 876.9 + 725.2 = 1602.1.$$

Degrees of freedom and mean squares:

$$df_{\text{Trt}} = a - 1 = 2, \quad df_E = N - a = 12, \quad df_T = N - 1 = 14,$$

$$MS_{\text{Trt}} = \frac{SS_{\text{Trt}}}{2} = 438.45, \qquad MS_E = \frac{SS_E}{12} = 60.4333.$$

The corresponding ANOVA table is

Source	df	SS	MS	F
Treatment	2	876.9	438.45	7.26
Error	12	725.2	60.43	
Total	14	1602.1		

The ANOVA F-statistic is

$$F = \frac{MS_{\text{Trt}}}{MS_E} = \frac{438.45}{60.4333} \approx 7.26,$$

which is compared to an F distribution with $(2, 12)$ degrees of freedom. The corresponding p-value from R is

$$p = 0.0086.$$

Since $p < 0.05$, we reject H_0 and conclude that there are significant differences in average starting salaries among the three institution types.

(c) Estimate of the standard deviation of the error

The error mean square MS_E is the usual unbiased estimator of σ^2. Hence

$$\hat{\sigma}^2 = MS_E = 60.4333, \qquad \hat{\sigma} = \sqrt{MS_E} \approx 7.77,$$

so the estimated error standard deviation is about \$7,770.

(d) Pairwise comparisons using Tukey HSD and Scheffé

With a balanced design and common replication $n = 5$, the standard error for pairwise differences of treatment means is

$$SE(\bar{y}_i - \bar{y}_j) = \sqrt{\frac{2MS_E}{n}} = \sqrt{\frac{2 \times 60.4333}{5}} = \sqrt{24.1733} \approx 4.92.$$

The pairwise sample mean differences (treatment j minus treatment i) are

$$\bar{y}_{\text{Priv}} - \bar{y}_{\text{Pub}} = 17.4, \quad \bar{y}_{\text{Ch}} - \bar{y}_{\text{Pub}} = 2.7, \quad \bar{y}_{\text{Ch}} - \bar{y}_{\text{Priv}} = -14.7.$$

Tukey HSD. For $\alpha = 0.05$, $a = 3$, and $df_E = 12$, the studentized range critical value from R is

$$q_{0.05,3,12} = 3.7729,$$

so

$$HSD = q_{0.05,3,12}\sqrt{\frac{MS_E}{n}} = 3.7729\sqrt{\frac{60.4333}{5}} \approx 3.7729 \times 3.48 \approx 13.1.$$

A pair (i, j) is significantly different if $|\bar{y}_i - \bar{y}_j| > HSD$.

- Private vs. Public: $|\bar{y}_{\text{Priv}} - \bar{y}_{\text{Pub}}| = 17.4 > 13.1 \Rightarrow$ significant.
- Church vs. Public: $|\bar{y}_{\text{Ch}} - \bar{y}_{\text{Pub}}| = 2.7 < 13.1 \Rightarrow$ not significant.
- Church vs. Private: $|\bar{y}_{\text{Ch}} - \bar{y}_{\text{Priv}}| = 14.7 > 13.1 \Rightarrow$ significant.

Thus, under Tukey's method, Private institutions have significantly higher starting salaries than both Public and Church-affiliated institutions, while Public and Church-affiliated institutions do not differ significantly.

Scheffé method. For a one-way ANOVA with $a = 3$ groups, Scheffé's critical factor at $\alpha = 0.05$ is

$$k_S = (a-1)F_{0.05,\,a-1,\,df_E} = 2F_{0.05,2,12}.$$

From R,

$$F_{0.05,2,12} = 3.8853, \quad k_S = 2 \times 3.8853 \approx 7.77.$$

For a pairwise contrast $C = \mu_i - \mu_j$, the estimated contrast is $\hat{C} = \bar{y}_i - \bar{y}_j$ with

$$SE(\hat{C}) = SE(\bar{y}_i - \bar{y}_j) \approx 4.92,$$

and Scheffé's test rejects $H_0 : C = 0$ if

$$\frac{\hat{C}^2}{SE(\hat{C})^2} > k_S.$$

Using the R output,

$$T^2_{\text{Priv-Pub}} = 12.52,$$
$$T^2_{\text{Ch-Pub}} = 0.30,$$
$$T^2_{\text{Ch-Priv}} = 8.94,$$

so

$$T^2_{\text{Priv-Pub}} > 7.77, \quad T^2_{\text{Ch-Priv}} > 7.77, \quad T^2_{\text{Ch-Pub}} < 7.77.$$

Therefore, Scheffé's method also finds that Private vs. Public and Private vs. Church-affiliated are significantly different, while Public vs. Church-affiliated is not. In this example, Tukey and Scheffé give the same pairwise conclusions.

(e) Kruskal–Wallis test

A Kruskal–Wallis test with $a = 3$ groups has $df = a - 1 = 2$ and, under H_0, the test statistic is approximately χ^2_2-distributed. We are told that the Kruskal–Wallis test statistic is $H = 8.09$. The 5% upper critical value for χ^2_2 is about 5.99, and since $H = 8.09 > 5.99$ the Kruskal–Wallis test rejects H_0 at the 5% level. The corresponding p–value

$$p = P(\chi^2_2 \geq 8.09) \approx 0.017$$

is also below 0.05. Because Kruskal–Wallis does not require normality and uses only the ranks of the data, this provides a robust, nonparametric check that still indicates differences in the distributions of salaries among the three institution types. Thus our conclusion in part (b) does not change.

(f) Possible improved model

The current model treats only institution type as a factor and ignores other important sources of variability. An improved model might:

- include additional factors or covariates such as academic discipline, geographic region, or research vs. teaching focus, which are known to affect starting salaries;
- use blocking on departments, regions, or year of hire to control for nuisance variation via a randomized complete block design;
- consider transformations (e.g., modeling log-salary) if residual plots indicate skewness or nonconstant variance across institution types.

Such extensions would likely reduce unexplained variability and provide a more realistic assessment of the effect of institution type on starting salaries. In particular, the sample variances (72.8, 48.2, 60.3) already suggest some heterogeneity in variability across institution types, and the small per–group sample size ($n = 5$) limits power. Adding important predictors (discipline, region, year of hire) and possibly modeling log–salaries would address these issues directly.

Problem 2

Three groups ($\#1, \#2, \#3$) of students were subjected to different teaching techniques and tested at the end of the course. There are $n_i = 6$ students in each group, and the average scores are

$$\bar{y}_1 = 72.3, \quad \bar{y}_2 = 79.5, \quad \bar{y}_3 = 65.8.$$

The overall variance of all 18 scores is 84.03, and we are told that the error variance estimate from the ANOVA model is

$$MS_E = 57.84.$$

(a) Complete ANOVA table

The grand mean is

$$\bar{y}_{..} = \frac{72.3 + 79.5 + 65.8}{3} = 72.5333.$$

The treatment sum of squares is

$$SS_{\text{Trt}} = n\sum_{i=1}^{3}(\bar{y}_i - \bar{y}_{..})^2 = 6\left[(72.3-72.5333)^2 + (79.5-72.5333)^2 + (65.8-72.5333)^2\right] = 563.56.$$

We are given $MS_E = 57.84$. The error degrees of freedom are

$$df_E = N - a = 18 - 3 = 15,$$

so

$$SS_E = MS_E \times df_E = 57.84 \times 15 = 867.6.$$

Thus the total sum of squares is

$$SS_T = SS_{\text{Trt}} + SS_E = 563.56 + 867.6 = 1431.16.$$

If we compute SS_T directly from the reported overall variance via $SS_T = (N-1) \times 84.03$, we obtain 1428.51, which differs slightly from 1431.16 due to rounding in the reported group means. In what follows we treat the given $MS_E = 57.84$ and the corresponding SS_E as authoritative.

Degrees of freedom and mean squares:

$$df_{\text{Trt}} = a - 1 = 2, \quad df_E = 15, \quad df_T = N - 1 = 17,$$

$$MS_{\text{Trt}} = \frac{SS_{\text{Trt}}}{2} = 281.78, \qquad MS_E = 57.84.$$

The ANOVA F-statistic is

$$F = \frac{MS_{\text{Trt}}}{MS_E} = \frac{281.78}{57.84} \approx 4.87.$$

The ANOVA table is therefore

Source	df	SS	MS	F
Treatment	2	563.56	281.78	4.87
Error	15	867.60	57.84	
Total	17	1431.16		

(b) Conclusion about differences in teaching techniques

We test

$$H_0 : \mu_1 = \mu_2 = \mu_3 \quad \text{vs} \quad H_a : \text{at least one } \mu_i \text{ differs.}$$

From the ANOVA table,

$$F = 4.8717, \quad df_1 = 2, \quad df_2 = 15.$$

The corresponding p-value from R is

$$p = 0.0234.$$

Since $p < 0.05$, we reject H_0 at the 5% level and conclude that there are significant differences in mean test scores among the three teaching techniques.

(c) Comparison of treatments to the control (group 2)

Suppose group 2 is the control and groups 1 and 3 are treatments (the problem statement appears to say "#1 and #2 are treatments," but since group 2 is designated as the control, we interpret this as a typo and treat #1 and #3 as the treatment groups). The differences in sample means relative to the control are

$$\bar{y}_1 - \bar{y}_2 = 72.3 - 79.5 = -7.2, \qquad \bar{y}_3 - \bar{y}_2 = 65.8 - 79.5 = -13.7.$$

The standard error for the difference between a treatment mean and the control mean (with equal group sizes $n = 6$) is

$$SE_{\text{diff}} = \sqrt{\frac{2MS_E}{n}} = \sqrt{\frac{2 \times 57.84}{6}} = 4.3909.$$

Thus the t-statistics (with $df = 15$) are

$$t_{1-2} = \frac{-7.2}{4.3909} = -1.64, \qquad t_{3-2} = \frac{-13.7}{4.3909} = -3.12.$$

The corresponding two–sided p–values (with 15 degrees of freedom) are approximately

$$p_{1-2} \approx 0.12, \qquad p_{3-2} \approx 0.007.$$

Thus, at $\alpha = 0.05$, only group 3 differs significantly from the control; group 1 is not significantly different from group 2.

Because the primary interest is in comparing each treatment to a single control, Dunnett's multiple-comparison procedure is appropriate. For $a = 3$ groups (two treatments and one control), $df_E = 15$, and familywise $\alpha = 0.05$, the Dunnett critical value is approximately

$$t_D \approx 2.40.$$

Using the same standard error $SE_{\text{diff}} = 4.3909$, the standardized differences are

$$t_{1-2} = -1.64, \qquad t_{3-2} = -3.12.$$

Since $|t_{3-2}| = 3.12 > 2.40$ while $|t_{1-2}| = 1.64 < 2.40$, Dunnett's test again concludes that group 3 has a significantly lower mean score than the control, whereas group 1 does not differ significantly from the control.

(d) Average treatment effect vs control with Bonferroni correction

We now test whether the *average* of the two treatment groups differs from the control. Consider the contrast

$$C = \tfrac{1}{2}\mu_1 + \tfrac{1}{2}\mu_3 - \mu_2.$$

Its estimate is

$$\hat{C} = \tfrac{1}{2}(72.3 + 65.8) - 79.5 = \tfrac{1}{2}(138.1) - 79.5 = 69.05 - 79.5 = -10.45.$$

For equal group sizes $n = 6$, the variance of $\hat{C}$ is

$$\operatorname{Var}(\hat{C}) = MS_E \left(\frac{0.5^2}{n} + \frac{0.5^2}{n} + \frac{1^2}{n} \right) = MS_E \cdot \frac{1.5}{n},$$

so

$$SE_C = \sqrt{\frac{1.5\, MS_E}{n}} = \sqrt{\frac{1.5 \times 57.84}{6}} = 3.8026.$$

The corresponding t-statistic (with $df = 15$) is

$$t = \frac{\hat{C}}{SE_C} = \frac{-10.45}{3.8026} = -2.75.$$

We are asked to test this contrast using a Bonferroni correction assuming a total of $K = 10$ post-hoc tests. With overall $\alpha = 0.05$, the Bonferroni-adjusted per-test level is

$$\alpha^* = \frac{\alpha}{K} = \frac{0.05}{10} = 0.005.$$

For a two-sided test, we use $\alpha^*/2 = 0.0025$ in each tail. The critical t value from R is

$$t_{1-\alpha^*/2,15} = t_{0.9975,15} = 3.286.$$

Since

$$|t| = 2.75 < 3.286,$$

we fail to reject $H_0 : C = 0$ at the Bonferroni-adjusted level. Thus, when controlling for a family of 10 post-hoc tests, the average of the two treatment groups is not significantly different from the control group, even though the unadjusted t-statistic suggests a noticeable decrease in score. Practically, the estimated average treatment effect of -10.45 points is fairly large, but the very conservative Bonferroni adjustment inflates the critical value, so this contrast does not reach statistical significance at the familywise 5% level.

Problem 3 (DAE 4.17)

The response is gene expression measured for three leukemia treatments: MP only, MP + HDMTX, and MP + LDMTX, with $n = 10$ observations per treatment ($N = 30$). The data are in `Mid1Prob3.csv`.

(a) Test whether treatment means differ (raw data)

A one-way fixed-effects ANOVA model is

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \quad i = 1, 2, 3,\ j = 1, \ldots, 10, \qquad \varepsilon_{ij} \sim \text{NID}(0, \sigma^2),$$

where i indexes treatment.

Sample means and standard deviations by treatment are

Treatment	n	Mean	SD
MP + HDMTX	10	479.3	371.0
MP + LDMTX	10	165.0	126.0
MP Only	10	238.0	308.0

From the ANOVA on the raw responses (R command `aov(Gene.Expression ~ Treatment)`),

Source	Df	SS	MS	F
Treatment	2	538,442	269,221	3.25
Error	27	2,236,974	82,851	

with p–value $p = 0.0544$.

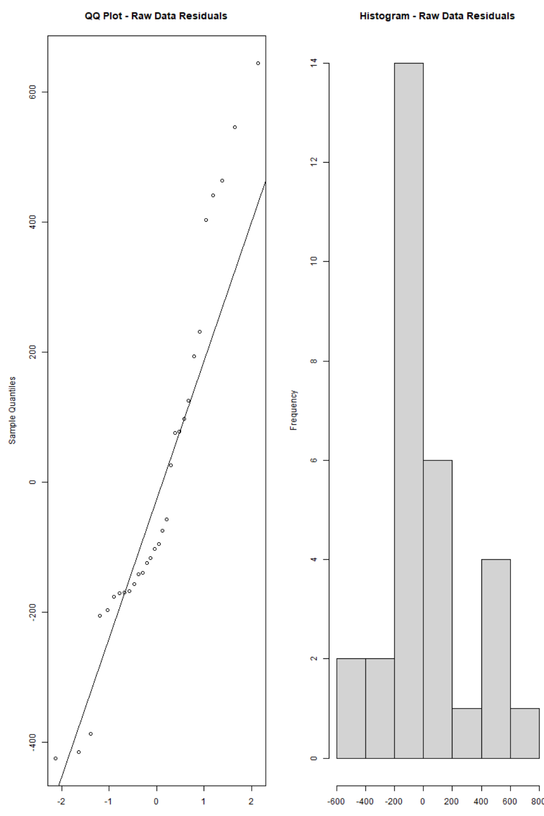
At $\alpha = 0.05$, p is slightly larger than 0.05, so we *do not* reject H_0 at the 5% level based strictly on this ANOVA. The evidence for a difference in mean gene expression across treatments is marginal but not conventionally significant.

(b) Normality check for raw data

The residuals from the raw-data ANOVA show substantial deviation from normality. The Shapiro–Wilk test applied to the raw residuals gives

$$W = 0.9165, \quad p = 0.0218,$$

which is below 0.05, indicating a statistically significant departure from normality. The residual histogram and QQ plot exhibit pronounced right skew and a few large positive residuals, consistent with the presence of outliers or heavy right tails. Thus the normality assumption is questionable on the original scale.



(c) Log transform and ANOVA on transformed data

Because the raw data are positive and strongly right-skewed, a log transform is natural. Let

$$Z_{ij} = \log(Y_{ij}),$$

and fit the one-way ANOVA model

$$Z_{ij} = \mu^* + \tau_i^* + \varepsilon_{ij}^*.$$

On the log scale, the sample means and standard deviations are

Treatment	n	Mean	SD
MP + HDMTX	10	5.73	1.13
MP + LDMTX	10	4.74	1.00
MP Only	10	4.79	1.18

The ANOVA on the log-transformed responses (R command `aov(logGE ~ Treatment)`) yields

Source	Df	SS	MS	F
Treatment	2	6.297	3.1485	2.57
Error	27	33.069	1.2248	

with $p = 0.0951$.

Thus, on the log scale the evidence for treatment differences is weaker: at $\alpha = 0.05$ we again fail to reject H_0; the p–value is between 0.05 and 0.10, so one might describe this as only marginal evidence of differences in mean log gene expression.

(d) Residual analysis for the transformed model

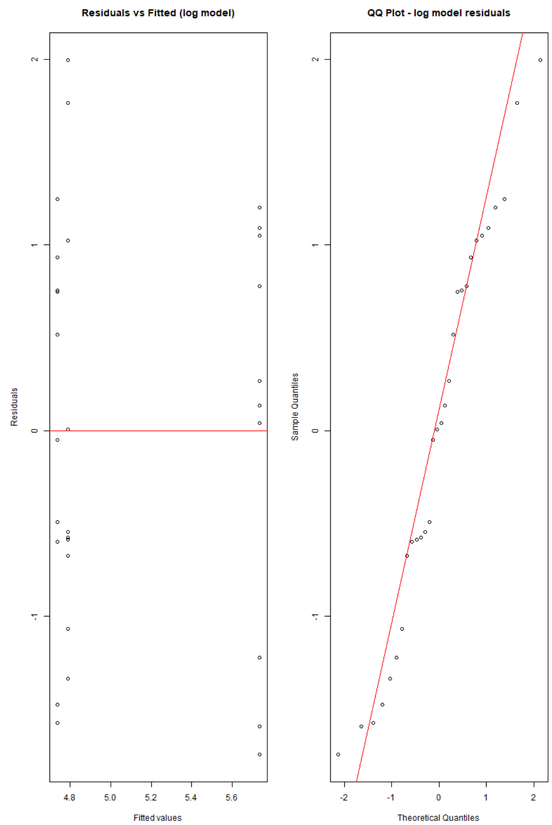
For the log-scale ANOVA, we examine the residuals. The residuals vs fitted plot for the log model shows a fairly symmetric cloud of points with no strong funnel shape; the variability appears roughly constant across fitted values. The QQ plot of the log-model residuals lies much closer to a straight line than for the raw data.

The Shapiro–Wilk test on the log-model residuals gives

$$W = 0.9565, \quad p = 0.2511,$$

which is not significant; we do not reject normality at the 5% level. This suggests that the log transform greatly improves the normality of the errors and makes the ANOVA assumptions more plausible.

In summary, the log transformation yields a model with much better residual behavior (approximately normal and homoscedastic), but even on this scale the treatment effect is not statistically significant at the $\alpha = 0.05$ level.



Residuals vs Fitted (log model)
Residuals
Fitted values
4.8
5.0
5.2
5.4
5.6
-1
0
1
2
QQ Plot - log model residuals
Sample Quantiles
Theoretical Quantiles
-2
-1
0
1
2

R Code

```
# Problem 1
means  <- c(Public = 53.4, Private = 70.8, Church = 56.1)
vars   <- c(Public = 72.8, Private = 48.2, Church = 60.3)
n      <- 5
grand_mean <- mean(means)
SS_trt <- n * sum((means - grand_mean)^2)
SS_E   <- sum((n - 1) * vars)
df_trt <- 2; df_E <- 12
MS_trt <- SS_trt / df_trt; MS_E <- SS_E / df_E
F_stat <- MS_trt / MS_E
p_val  <- 1 - pf(F_stat, df_trt, df_E)
sigma_hat <- sqrt(MS_E)
SE_diff <- sqrt(2 * MS_E / n)
diffs <- c(
  Priv_Pub = means["Private"] - means["Public"],
  Ch_Pub   = means["Church"]  - means["Public"],
  Ch_Priv  = means["Church"]  - means["Private"]
)
qcrit <- qtukey(0.95, nmeans = 3, df = df_E)
HSD   <- qcrit * sqrt(MS_E / n)
Fcrit <- qf(0.95, df1 = df_trt, df2 = df_E)
k_S   <- (3 - 1) * Fcrit
T2    <- (diffs^2) / (SE_diff^2)

# Problem 2
means_P2 <- c(G1 = 72.3, G2 = 79.5, G3 = 65.8)
n_P2     <- 6
a_P2     <- 3
N_P2     <- a_P2 * n_P2
overall_var <- 84.03
MS_E_P2     <- 57.84
grand_mean_P2 <- mean(means_P2)
SS_T_P2 <- overall_var * (N_P2 - 1)
SS_trt_P2 <- n_P2 * sum((means_P2 - grand_mean_P2)^2)
df_trt_P2 <- a_P2 - 1
df_E_P2   <- N_P2 - a_P2
SS_E_P2   <- MS_E_P2 * df_E_P2
SS_T_P2_consistent <- SS_trt_P2 + SS_E_P2
MS_trt_P2 <- SS_trt_P2 / df_trt_P2
F_stat_P2 <- MS_trt_P2 / MS_E_P2
p_val_P2  <- 1 - pf(F_stat_P2, df_trt_P2, df_E_P2)
```

```
SE_diff_P2 <- sqrt(2 * MS_E_P2 / n_P2)
diffs_vs_control <- c(
  G1_minus_G2 = means_P2["G1"] - means_P2["G2"],
  G3_minus_G2 = means_P2["G3"] - means_P2["G2"]
)
t_stats_vs_control <- diffs_vs_control / SE_diff_P2

C_hat <- 0.5 * means_P2["G1"] + 0.5 * means_P2["G3"] - means_P2["G2"]
SE_C  <- sqrt(MS_E_P2 * 1.5 / n_P2)
t_C   <- C_hat / SE_C
alpha      <- 0.05
K          <- 10
alpha_star <- alpha / K
t_crit_Bonf <- qt(1 - alpha_star / 2, df = df_E_P2)

# Problem 3
dat <- read.csv("Stats 490/Exam1/Mid1Prob3.csv")
dat$Treatment <- factor(dat$Treatment)

library(dplyr)
group_stats <- dat %>%
  group_by(Treatment) %>%
  summarise(
    n    = n(),
    mean = mean(Gene.Expression),
    sd   = sd(Gene.Expression)
  )

fit_raw <- aov(Gene.Expression ~ Treatment, data = dat)
anova_raw <- summary(fit_raw)[[1]]
res_raw <- residuals(fit_raw)

par(mfrow = c(1, 2))
qqnorm(res_raw, main = "QQ Plot - Raw Data Residuals")
qqline(res_raw)
hist(res_raw, main = "Histogram - Raw Data Residuals",
     xlab = "Residuals")
par(mfrow = c(1, 1))

shapiro_raw <- shapiro.test(res_raw)

dat$logGE <- log(dat$Gene.Expression)

group_stats_log <- dat %>%
```

```r
  group_by(Treatment) %>%
  summarise(
    n    = n(),
    mean = mean(logGE),
    sd   = sd(logGE)
  )

fit_log  <- aov(logGE ~ Treatment, data = dat)
anova_log <- summary(fit_log)[[1]]
res_log    <- residuals(fit_log)
fitted_log <- fitted(fit_log)

par(mfrow = c(1, 2))
plot(fitted_log, res_log,
     main = "Residuals vs Fitted (log model)",
     xlab = "Fitted values", ylab = "Residuals")
abline(h = 0, col = "red")
qqnorm(res_log, main = "QQ Plot - log model residuals")
qqline(res_log, col = "red")
par(mfrow = c(1, 1))

shapiro_log <- shapiro.test(res_log)
```